

6 (a) ${}^1_1\text{H}$ (proton)

(b) (i) $\Delta m = (17.004507 \text{ u} + 1.008142 \text{ u}) - (14.007525 \text{ u} + 4.003860 \text{ u})$

$$= 1.264 \times 10^{-3} \text{ u}$$

$$= 2.098 \times 10^{-30} \text{ kg}$$

The change in rest-mass energy

$$E = \Delta mc^2 = (2.098 \times 10^{-30})(3.00 \times 10^8)^2$$

$$= 1.9 \times 10^{-13} \text{ J}$$

(ii) The combined rest-mass of the products is greater than that of the reactants. This means that energy has to be supplied to enable the reaction to occur.

The energy is supplied from the kinetic energy of the α -particle, which is converted into mass through mass-energy equivalence $E = \Delta mc^2$

(iii) The fact that the products possess kinetic energy means that the kinetic energy of the α -particle has to be greater than the increase in rest-mass energy ($1.9 \times 10^{-13} \text{ J}$).

The kinetic energy of the α -particle has to be equal to $1.9 \times 10^{-13} \text{ J}$ + total kinetic energy of the oxygen and proton after the reaction.

The momentum of the system is conserved too. Hence the velocities of the products (including their directions) depend on the velocities of the reactants.

Note: Some students may not realise that the reaction produced an increase in mass.

7 (a) (i) Work done W by a force F is the product of the force and the distance s moved in